

Project and Presentation Rubric for 2026 NACME NVB Engineering Design Project

Total: 100 pts

1) Problem Framing & Impact (10 pts)

Focus: Does the team clearly define the problem and why it matters?

Excellent (9-10)	Proficient (7-8)	Developing (4-6)	Insufficient (0-3)
- Clearly defines user needs, use case, and engineering problem - Demonstrates strong real-world relevance and impact - Articulates clear success criteria	- Problem and use case identified - Some connection to impact - Minor gaps in clarity or specificity	- Partial or vague problem framing - Limited connection to real-world use or stakeholders	Unclear or missing problem definition

2) Technical Implementation & Data Analytics (25 pts)

Focus: Correctness, depth, and quality of data analysis and app functionality

Excellent (22-25)	Proficient (18-21)	Developing (12-17)	Insufficient (0-11)
- Correct and well-justified data processing, metrics, and models - Effective use of sensor data (.mat files), including cleaning and transformation - Fully functional app with meaningful interactivity	- Minor errors - App functional - Data processing mostly appropriate	- Noticeable issues in implementation - Limited functionality or weak analysis	- Major errors - Incomplete or non-functional solution

• Strong integration between data, analysis, and UI			
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3) Analysis, Validation & Engineering Reasoning (15 pts)

Focus: Evidence-based reasoning and validation of results

Excellent (14-15)	Proficient (11-13)	Developing (7-10)	Insufficient (0-6)
• Results are validated with checks (e.g., sanity checks, comparisons, edge cases) • Assumptions are clearly stated and justified • Demonstrates strong engineering reasoning and interpretation	• Some validation and reasoning • Minor gaps	• Limited validation • Weak explanation of results	• No validation or reasoning

4) App Design, Visualization & User Experience (15 pts)

Focus: Quality and clarity of the app interface and data communication

Excellent (14-15)	Proficient (11-13)	Developing (7-10)	Insufficient (0-6)
• Clear, intuitive UI with effective layout • High-quality visualizations (well-labeled, meaningful, accurate) • Includes interactive features (buttons, inputs) and visual feedback (alerts/indicators) • Designed for a non-expert user	• Functional UI with generally clear visuals • Minor usability issues	• Basic UI • Unclear or incomplete visualizations	• Poor or missing UI/visualization

5) Code Quality, Reproducibility & Documentation (10 pts)

Focus: Professional software practices and project organization

Excellent (9-10)	Proficient (7-8)	Developing (4-6)	Insufficient (0-3)
• Clean, well-organized code with comments • GitHub repository structured and complete • README clearly explains setup, data, and how to run the app • Results reproducible from provided materials	• Mostly complete • Minor gaps in documentation or organization	• Disorganized or partially reproducible	• Missing key components • Not reproducible

6) Communication, Storytelling & Professionalism (15 pts)

Focus: Effectiveness of presentation (Shark Tank-style)

Excellent (14-15)	Proficient (11-13)	Developing (7-10)	Insufficient (0-6)
• Clear, engaging introduction of problem and solution • Compelling story (motivation, inspiration, and design rationale) • Explains project design, functionality, and impact in a structured, logical way • Confident delivery: strong eye contact, clear voice, appropriate pacing • Professional demeanor (appearance, tone, preparation) • Visual materials enhance understanding • Presentation is polished, persuasive, and audience-focused	• Clear explanation of project; some storytelling elements • Generally professional delivery • Minor issues with clarity or engagement	• Basic explanation • Limited storytelling or structure • Uneven delivery (e.g., reliance on notes, unclear speaking) • Limited professionalism	• Unclear or incomplete explanation • Lack of preparation or professionalism • Difficult to follow presentation

7) Design Strategy, Innovation & Persuasiveness (10 pts)

Focus: Strength of engineering design thinking and originality

Excellent (9-10)	Proficient (7-8)	Developing (4-6)	Insufficient (0-3)
• Creative and innovative solution • Clearly explains design decisions and tradeoffs • Demonstrates strong differentiation and value • Persuasively defends design choices	• Solid design with some explanation of decisions	• Limited innovation • Weak design justification	• Minimal design thinking or originality